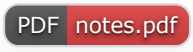
Notes_25.06: Naive Bayes Example & Laplace Smoothing







🌐 **Language: English** | हिंदी / Hinglish | ⬅️ Lecture 25 Overview

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📌 Executive Summary

Step-by-step tabular calculation example and additive Laplace smoothing to solve zero probability.

🎯 Mathematical Formulations

Zero Frequency Problem & Additive Laplace Smoothing ($\alpha = 1$):

$$P(w_i \mid y) = \frac{\text{Count}(w_i, y) + \alpha}{\sum_{w \in V} \text{Count}(w, y) + \alpha|V|}$$

Jahan $|V|$ vocabulary size (unique words ki sankhya) hai aur $\alpha = 1$ smoothing parameter hai.

💡 Conceptual Principles & Architecture

Zero Frequency Problem Kya Hai?

Maan lijiye training data mein word `"Bitcoin"` kabhi Ham email mein nahi aaya. Toh $P(\text{"Bitcoin"} \mid \text{Ham}) = 0$. Jab test email aayegi jisme `"Bitcoin"` likha hoga, toh probability multiply hokar: $P(\mathbf{x} \mid \text{Ham}) = P(x_1 \mid \text{Ham}) \times \cdots \times 0 = 0$! Ek naye word ki wajah se poori probability zero ho gayi!

Solution: Laplace Smoothing

Hum numerator mein $+1$ aur denominator mein $+|V|$ add kar dete hain. Isse kisi bhi word ki probability kabhi exactly zero nahi hoti.

💻 Python Production Implementation

```
# Laplace smoothing example
count_word = 0
total_words_in_class = 500
vocab_size = 1000
alpha = 1.0

prob_without_smoothing = count_word / total_words_in_class
prob_with_smoothing = (count_word + alpha) / (total_words_in_class + alpha * voca

print(f"Without smoothing : {prob_without_smoothing}")
print(f"With smoothing    : {prob_with_smoothing:.6f}")
```